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Total No. of Printed Pages : 2

Roll No

AU/ME-228-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Machine Design - I

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat sketch, if required.

1. a) State the maximum shear strain theory and draw its graph.
b) Define stress concentration. What are the causes of stress concentration?
2. a) What are different modes of failure? Explain.
b) What do you mean by Fatigue?
3. a) Discuss the modified Goodman diagram's uses.
b) Discuss the role of Fasteners and Keys.
4. a) Discuss the power screws use in brief.
b) The shear strength, tensile strength & compressive strength of a rivet joint are 100N, 120N and 150N respectively. If strength of unriveted plate is 200N. Find the efficiency of rivet joint.

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5. Derive an expression for equivalent stiffness of two springs connected, having stiffness K_1 and K_2 .
 - i) In series
 - ii) In parallel
6.
 - a) Write design steps for bolted joints.
 - b) What do you mean by stable and unstable operation?
7.
 - a) What are the criteria for selection of ball and roller bearings?
 - b) Write short note on boundary lubrication.
8.
 - a) Differentiate fillet and butt welds with neat sketches.
 - b) Write short notes on any two of the following.
 - i) Cyclic loading
 - ii) Surface factor
 - iii) Size factor
 - iv) Loading factor
